

**Section A**

Answer all questions.

1. Lead is a metal that can bind to the –SH groups of enzyme molecules.

- (a) (i) Name the bond in an enzyme molecule that is formed from –SH [1]  
groups.

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- (ii) Suggest how an enzyme molecule may be affected when lead ions bind to it. [3]

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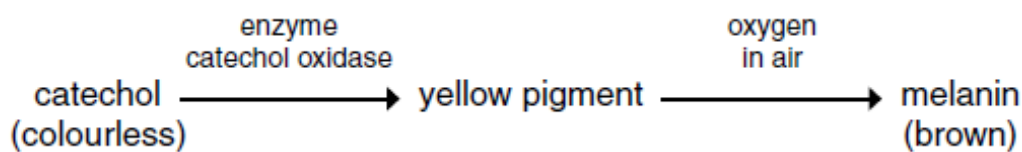
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- (b) When an apple is cut and exposed to the air, the cut surface will turn brown after a few minutes. This is the result of reactions involving the enzyme catechol oxidase.



- (i) Suggest why tinned apple does not turn brown. [1]

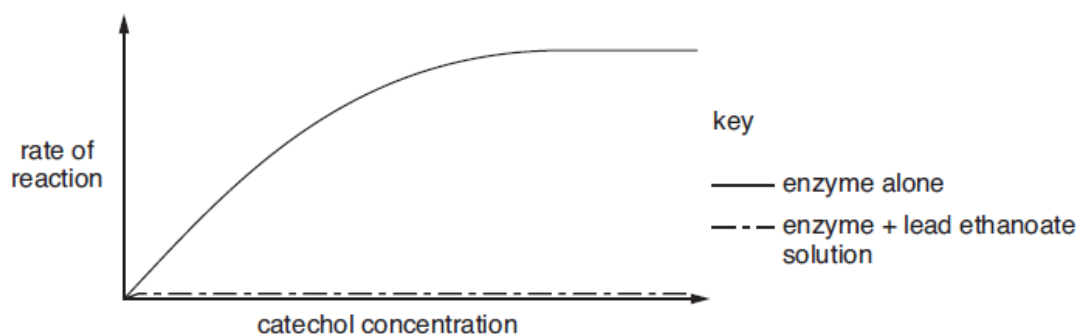
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The rate of reaction of catechol oxidase was investigated for a range of concentrations of catechol solution.

The experiment was then repeated for the same concentrations of catechol solution, but with the addition of a fixed volume and concentration of lead ethanoate solution.

The results are shown in Fig. 1.1.



**Fig 1.1**

- (ii) What conclusions may be drawn about the way in which lead ethanoate solution affects the activity of the enzyme catechol oxidase?

Use the information provided in **(a)** and **(b)** to help you.

Give reasons for your conclusions.

[4]

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[Total 9]

2. (a) Down's syndrome in humans usually results from acquiring an extra chromosome 21 from the mother.

Explain how the presence of an extra chromosome; such as chromosome 21, occurs. [3]

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- (b) A small section of chromosome 21, the so-called Down's syndrome critical region (DSCR) causes the abnormalities associated with the syndrome.

Precise regulation of gene expression by a particular group of transcription factors is essential for normal development. In inactive cells the transcription factors are found in the cytoplasm, but in active cells, they move into the nucleus.

In humans, a gene, **D**, has been identified in the DSCR that affects a transcription factor. Mice that have been genetically engineered to over-express **D** show many of the symptoms of Down's syndrome.

The nuclear / cytoplasmic ratio of the transcription factors and the expression of the genes that they control were measured in the genetically engineered mice and in normal mice. The results are shown in Table 2.1.

**Table 2.1**

|                                                                        | Normal mice | Mice engineered to over-express <b>D</b> |
|------------------------------------------------------------------------|-------------|------------------------------------------|
| Nuclear / cytoplasmic ratio of transcription factors                   | 2.75        | 0.75                                     |
| Relative expression of genes controlled by these transcription factors | ×12         | ×1                                       |

With reference to the information provided in **(b)**,

- (i) describe the effect of the over-expression of the human gene, **D**, in mice; [2]

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- (ii) suggest why the genetically engineered mice show many of the symptoms of Down's syndrome. [3]

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[Total 8]

3. *E. coli* proliferates faster on the monosaccharide glucose than it does on the disaccharide lactose for two reasons: (1) lactose is taken up more slowly than glucose and (2) lactose must be hydrolysed to glucose and galactose (by  $\beta$ -galactosidase) before it can be further metabolised.

When *E. coli* is grown on a medium containing a mixture of glucose and lactose, it proliferates with complex kinetics (Fig. 3.1). The bacteria proliferate faster at the beginning than at the end, and there is a lag between these two phases when they virtually stop dividing. Assays of the concentrations of the two sugars in the medium show that glucose falls to very low levels after a few cell doublings (Fig. 3.1, circles), but lactose remains high until the end of the experimental time course (not shown).

Although the concentration of lactose is high throughout most of the experiment,  $\beta$ -galactosidase, which is regulated as part of the *Lac* operon, is not induced until more than 100 minutes have passed (Fig. 3.1, triangles).

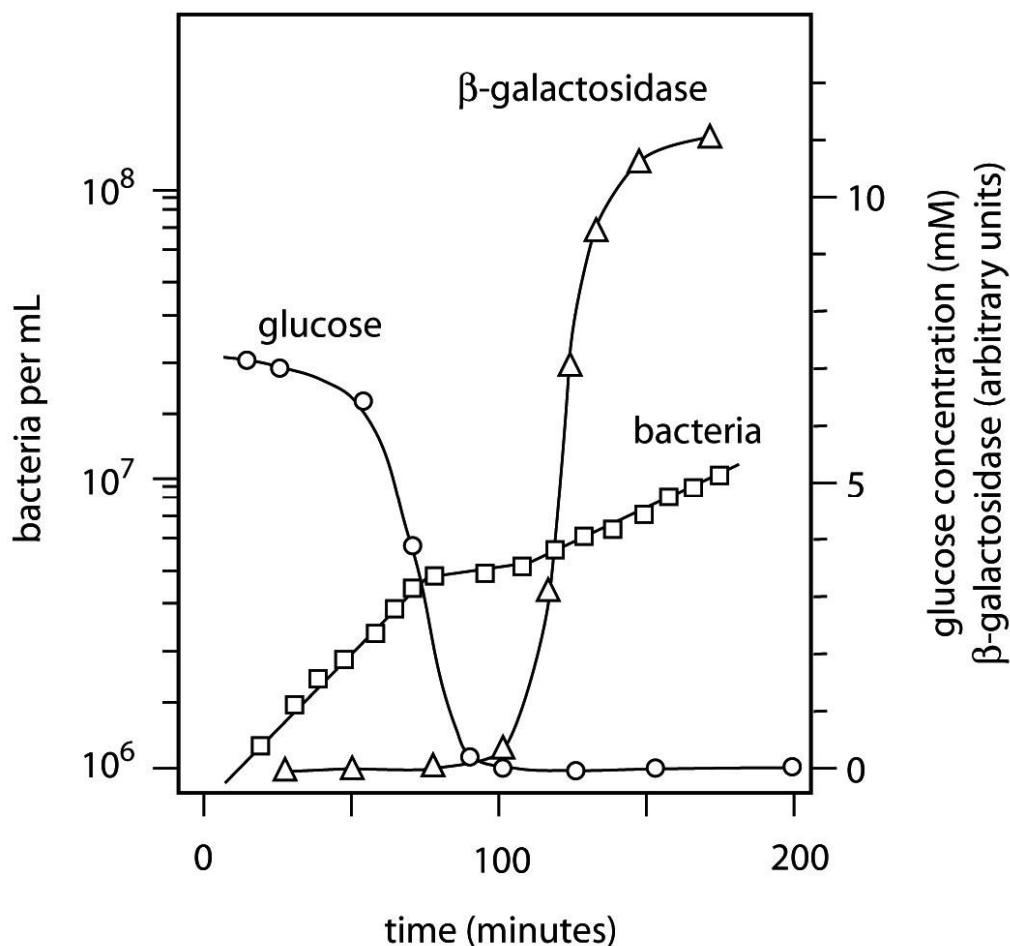


Fig. 3.1

- (a) Explain the kinetics of bacterial proliferation during the experiment: account for the rapid initial rate, the slower final rate, and the delay in the middle of the experiment. [3]

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- (b) Explain why the *Lac* operon is not induced by lactose during the rapid initial phase of bacterial proliferation. [5]

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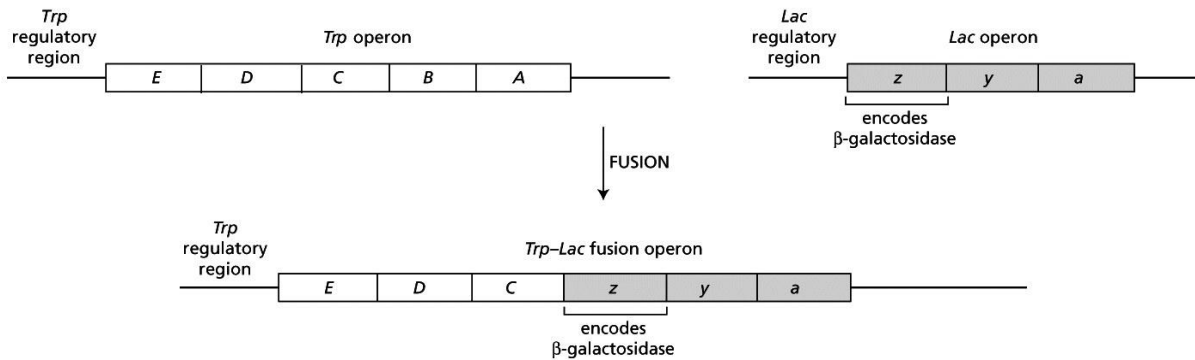
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Imagine that you have created a fusion between the *Trp* operon, which encodes the enzymes for tryptophan biosynthesis, and the *Lac* operon, which encodes the enzymes necessary for lactose utilisation (Fig. 3.2).



**Fig. 3.2**

(c) Under which set of conditions (**A-F**) below will  $\beta$ -galactosidase be expressed in the strain of that carries the fused operon shown in Fig. 3.2? Explain your answer.

- A.** Only when lactose and glucose are both absent.
- B.** Only when lactose and glucose are both present.
- C.** Only when lactose is absent and glucose is present.
- D.** Only when lactose is present and glucose is absent.
- E.** Only when tryptophan is absent.
- F.** Only when tryptophan is present.

[3]

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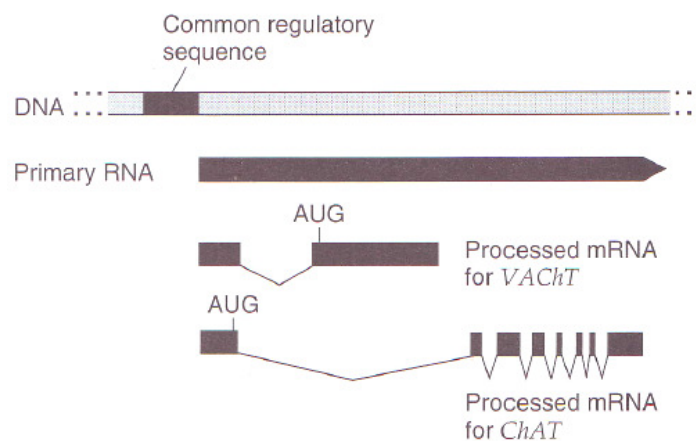


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Although eukaryotes lack operons like those found in prokaryotes, the exceptional conserved organisation of the *ChAT* / *VACht* locus is reminiscent of a prokaryotic operon.

*ChAT* is the gene for the enzyme choline acetyltransferase, which synthesises acetylcholine, a neurotransmitter released by one neuron to signal another neuron. *VACht* is the gene for the vesicular acetylcholine transporter protein, which packages acetylcholine into vesicles before its release from a neuron. Both *ChAT* and *VACht* are expressed in the same neuron.

Part of the *VACht* gene is nested within the first intron of the *ChAT* gene, and the two genes share a common regulatory protein and a first exon. The structure of a primary mRNA and two processed mRNA transcripts produced by this locus are shown in Fig. 3.3. The common regulatory region important for transcription of the locus in neurons is shown in the DNA, black triangles in RNA represent exons, lines connecting the exons represent spliced intron regions, and AUG indicates the translation start codons within the *ChAT* and *VACht* mRNAs. Polyadenylation sites are not shown.



**Fig. 3.3**

- (d) In what ways is the organisation of the *VACht* / *ChAT* locus reminiscent of a bacterial operon? [2]

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- (e) Why is the organization of this locus not structurally equivalent to a bacterial operon? [2]

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- (f) Based on the transcripts shown, identify one mode of regulation that might be used to obtain two different protein products from the single primary mRNA. [1]

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[Total 16]

4. (a) Wild-type fruit flies, *Drosophila melanogaster*, have red eyes. The red pigment, drospterin, is derived from the amino acid tryptophan by a series of steps, each controlled by a different gene.

Flies that were each heterozygous for two of the genes, **A/a** and **B/b**, were mated. The offspring had three different eye colours in a ratio of 9 red-eyed flies: 3 brown-eyed flies: 4 cinnabar-eyed flies. There were approximately equal numbers of male and female flies with each phenotype.

The genotypes associated with each phenotype are shown in the Punnett square of the cross in Table 4.1.

**Table 4.1**

|                            |           | female parent <b>AaBb</b>             |                    |                       |                       |
|----------------------------|-----------|---------------------------------------|--------------------|-----------------------|-----------------------|
|                            |           | genotypes and phenotypes of offspring |                    |                       |                       |
| male parent<br><b>AaBb</b> | gametes   | <b>AB</b>                             | <b>Ab</b>          | <b>aB</b>             | <b>ab</b>             |
|                            | <b>AB</b> | AABB<br>red eyes                      | AABb<br>red eyes   | AaBB<br>red eyes      | AaBb<br>red eyes      |
|                            | <b>Ab</b> | AABb<br>red eyes                      | AAbb<br>brown eyes | AaBb<br>red eyes      | Aabb<br>brown eyes    |
|                            | <b>aB</b> | AaBB<br>red eyes                      | AaBb<br>red eyes   | aaBB<br>cinnabar eyes | aaBb<br>cinnabar eyes |
|                            | <b>ab</b> | AaBb<br>red eyes                      | Aabb<br>brown eyes | aaBb<br>cinnabar eyes | aabb<br>cinnabar eyes |

With reference to Table 4.1 and the information provided,

- (i) state precisely the name given to this type of interaction between genes; [1]

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- (ii) describe the effects of the two genes on eye colour in *D. melanogaster*; [4]

gene **A/a**

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*gene B/b*

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- (iii) explain whether or not there is any evidence from the 9 : 3 : 4 ratio of phenotypes that the two genes, **A/a** and **B/b**, are on the same chromosome; [3]

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- (iv) assuming that the two genes code for enzymes involved in drosopterin production, suggest an explanation for the gene interaction in the pathway of drosopterin production. [3]

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- (b) The expected results from the Mendelian dihybrid cross described in (a) would be **four** different phenotypes in a ratio of 9: 3: 3: 1.

The phenotypes of 160 randomly chosen offspring from this cross were recorded. The expected and actual numbers of flies with each phenotype are shown in Table 4.2.

**Table 4.2**

| Phenotypes       | Number of flies with each phenotype |            |               |                       |
|------------------|-------------------------------------|------------|---------------|-----------------------|
|                  | Red eyes                            | Brown eyes | Cinnabar eyes | Another coloured eyes |
| Expected numbers | 90                                  | 30         | 30            | 10                    |
| Actual numbers   | 96                                  | 26         | 38            | 0                     |

A  $\chi^2$  (chi-squared) test was performed on these data to determine whether the results differed significantly from the expected Mendelian ratio.

The probability was found to be less than 0.01.

State what conclusions may be drawn from this probability. [3]

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[Total 14]

- 5 In muscle cells, adrenaline binds to the  $\beta$ -adrenergic receptor to initiate a signalling cascade that leads to the breakdown of glycogen (Fig. 5.1).

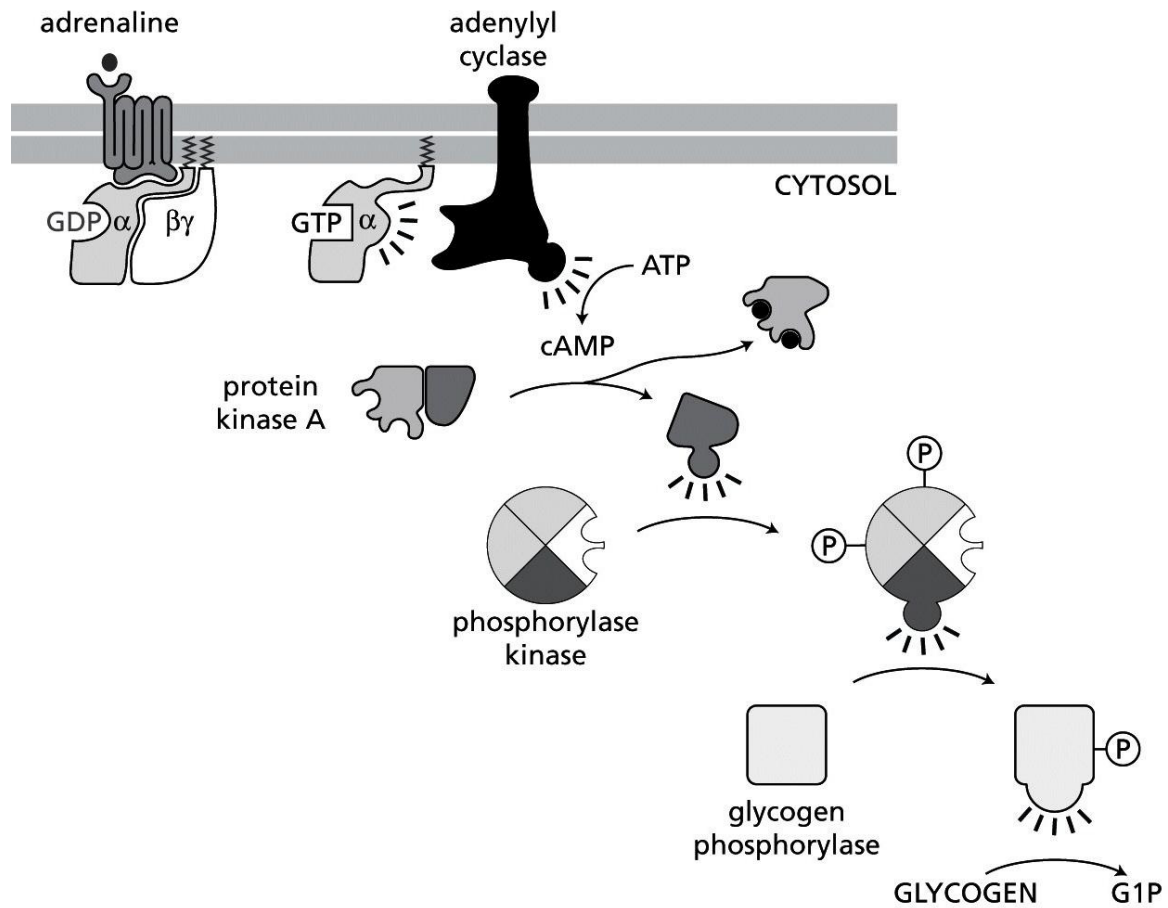


Fig. 5.1

- (a) Describe **two** points in this pathway in which the signal is amplified. [4]

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- (b)** A critical feature of all signaling cascades is that they must be turned off rapidly when the extracellular signal is removed.

Examine the signaling cascade in Fig. 5.1 and describe how each component of this signaling pathway is returned to its inactive state when adrenaline is removed. [5]

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- (c) During a marathon, runners draw heavily on their internal reserves of glycogen (carbohydrate) and triglycerides (fat) to fuel their muscle contraction. Initially, energy is derived mostly from carbohydrates, with increasing amounts of fats being used as the race progresses.

If runners use up their muscle glycogen reserves before they finish a race, they hit what is known as 'the wall', a point of diminished performance that arises because fatty acids from triglycerides breakdown cannot be delivered to the muscles quickly enough to sustain maximum effort. One trick that marathon runners use to avoid the wall is to drink a cup of strong black coffee an hour or so before the race begins. Coffee contains caffeine, an inhibitor of cyclic AMP phosphodiesterase.

How does inhibition of this enzyme help to avoid 'the wall'? [3]

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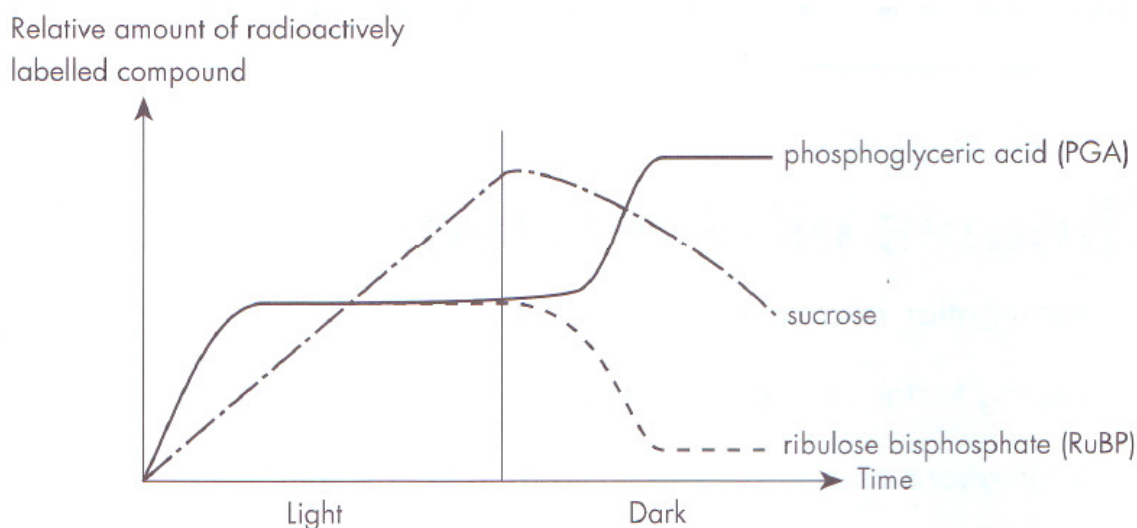
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[Total:12]

- 6 In an experiment to determine the biochemical pathway of photosynthesis, unicellular algae were allowed to photosynthesise actively with radioactive carbon dioxide ( $^{14}\text{CO}_2$ ) in the presence of light. After a certain period of time, the algae were left in the dark by switching off the light. Cells were taken out from time to time to extract the substances formed in them throughout the experimental period. The variation of the relative amount of radioactively labelled compound with time is shown in Fig. 6.1.



**Fig. 6.1**

- (a) What is the function of radioactive carbon dioxide in this experiment? [1]

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**(b)** Account for the changes in the amount of radioactive PGA with time during

**(i)** the light period and [4]

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**(ii)** during the dark period. [4]

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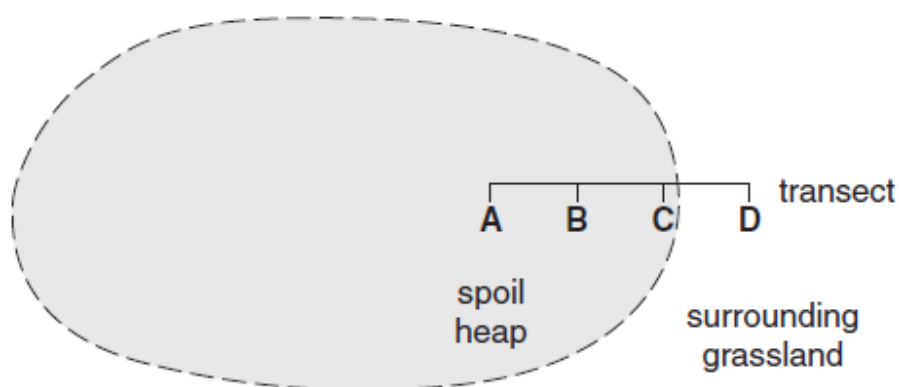
**(c)** Account for the drop in the amount of radioactive sucrose after the light was switched off. [1]

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[Total:10]

7. The spoil heaps of old copper mines are sparsely populated by plants because of the toxic levels of copper in the soil. However, copper-tolerant varieties of grasses such as the common Bent Grass, *Agrostis tenuis*, can become established in such habitats. The distribution of copper-tolerant varieties of *A. tenuis* was investigated on the spoil heap of an old copper mine, as shown in Fig. 7.1.



**Fig. 7.1**

The copper content of the soil and the copper-tolerance of the plants were measured at sites **A**, **B**, **C** and **D** along the transect.

Plants were taken from each of these sites and their copper-tolerance was determined by growing half the plants from each site in a culture solution without copper ions and the other half in a culture solution with a known concentration of copper ions.

The mean root growth of the plants in each solution was measured. The percentage reduction in mean root growth due to the presence of copper ions was calculated. The results are shown in Table 7.1.

The results show that the more copper-tolerant the plants, the smaller the percentage reduction in mean root growth.

**Table 7.1**

| Site along transect | Copper content of soil /<br>ppm | % reduction in mean root<br>growth |
|---------------------|---------------------------------|------------------------------------|
| <b>A</b>            | 750                             | 45                                 |
| <b>B</b>            | 350                             | 52                                 |
| <b>C</b>            | 125                             | 67                                 |
| <b>D</b>            | 35                              | 80                                 |

- (a) Suggest how copper ions reduce root growth in *A. tenuis*. [2]

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- (b) Explain how natural selection has resulted in the distribution of copper-tolerance in *A. tenuis* as shown in Fig. 7.1 and Table 7.1. [4]

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- (c) For the copper-tolerant varieties of *A. tenuis* to develop into separate species, the population must split into two separate gene pools.

- (i) Name the **two** types of isolating mechanism that can lead to speciation. [2]

1

2

(ii) The apple maggot fly, *Rhagoletis pomonella*, lives on hawthorn berries in North America.

- The larvae develop in the berries and, when adult, the females emerge and mate with males attracted from other hawthorn berries.
- The mated females then seek out a new hawthorn berry in which to lay their eggs.
- When apple trees were introduced to North America in the nineteenth century, some female flies switched from hawthorn to apple.
- Female flies always lay their eggs on the same type of plant from which they emerged.
- Flies emerge from apples one month before flies emerge from hawthorn berries.

Explain how the switch in host plant by some female flies could lead to speciation occurring.

[3]

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[Total 11]

**Section B**

Answer only **one** question

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where applicable.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8**    **(a)**    Explain how meiosis and fertilization in sexual reproduction lead to variation within a population. [6]
- (b)**    Outline the role of NAD in aerobic and anaerobic respiration. [6]
- (c)**    Relate the structure of the T<sub>4</sub> bacteriophage to its function. [8]
- 9**    **(a)**    Discuss the sickle cell mutation and its prevalence in malaria-stricken areas of Africa. [6]
- (b)**    Describe the process of chemiosmosis in the chloroplast and the processes that generate and maintain the proton gradient. [6]
- (c)**    Discuss mechanisms, other than mutation, that may increase the genetic variance / adaptability of bacteria cells (to its environment). [8]